

Eagle Trading Analyzer

A powerful Android trading analysis application that captures your screen in real-time and analyzes candlestick patterns with AI-powered insights. Features screen recording, pattern recognition, and intelligent trading signals.

Features

- **Real-Time Screen Capture:** Analyzes live trading charts from your screen
- **Candlestick Pattern Recognition:** Detects 8+ Japanese candlestick patterns
- **Indicator Score (2-5 Scale):** Proprietary scoring system for signal strength
- **Trading Signals:** BUY, SELL, or WAIT recommendations with confidence levels
- **Eagle Icon & Sound Effects:** Professional UI with audio alerts for strong signals
- **Customizable Settings:** Adjust capture interval, sensitivity, and notification preferences
- **Dark Modern UI:** Beautiful, intuitive interface optimized for trading
- **Android 8+ Support:** Compatible with Android 8 (API 26) and above

Requirements

- Android 8.0 (API level 26) or higher
- Screen recording permission
- Microphone permission (for sound alerts)

Installation

Option 1: Build from Source (Recommended)

1. **Clone or download** the project
2. **Open in Android Studio:**
 - File → Open → Select the project directory
3. **Sync Gradle** and wait for dependencies to download
4. **Build the APK:**
 - Build → Build Bundle(s) / APK(s) → Build APK(s)
5. **Install on device:**
 - Connect Android device via USB

- Run → Run 'app'

Option 2: Command Line Build

Bash

```
cd eagle-trading-analyzer
./gradlew assembleRelease
# APK will be generated in: app/build/outputs/apk/release/
```

How to Use

1. **Start the App:** Launch Eagle Trading Analyzer
2. **Grant Permissions:** Allow screen recording and audio permissions
3. **Press START:** Begin real-time analysis
4. **Select Recording Area:** Choose your trading chart area
5. **Monitor Signals:** Watch for BUY/SELL signals with strength indicators
6. **Customize:** Open Settings to adjust sensitivity and alerts

Architecture

Core Components

- **CandleAnalyzer.java:** Pattern recognition engine
 - Detects 8+ candlestick patterns (Doji, Hammer, Engulfing, etc.)
 - Calculates psychology scores (Bull/Bear)
 - Generates trading signals with 2-5 indicator score
- **ScreenCaptureService.java:** Screen recording service
 - Captures frames at configurable intervals
 - Extracts candlestick data using computer vision
 - Broadcasts signals to analysis service
- **AnalysisService.java:** Background analysis engine
 - Processes captured frames
 - Plays eagle sound for strong signals
 - Updates UI with real-time data

- **MainActivity.java:** Main UI controller
 - Manages start/stop controls
 - Displays current signals and status
 - Handles permission requests
- **SettingsActivity.java:** Configuration screen
 - Capture interval (500-10000ms)
 - Minimum signal strength (0-100%)
 - Sensitivity slider (0-100%)
 - Sound and notification toggles

Candlestick Patterns Recognized

1. **Doji** - Market indecision
2. **Gravestone Doji** - Bearish reversal signal
3. **Dragonfly Doji** - Bullish reversal signal
4. **Hammer** - Bullish reversal (after downtrend)
5. **Shooting Star** - Bearish reversal (after uptrend)
6. **Marubozu** - Strong directional movement
7. **Engulfing** - Reversal pattern
8. **Morning/Evening Star** - Multi-candle reversal
9. **Tweezer** - Support/resistance confirmation

Indicator Score System (2-5 Scale)

- **5:** Strong oversold (RSI < 30) - Excellent buy signal
- **4:** Moderate oversold (RSI 30-40) - Good buy signal
- **3:** Neutral (RSI 40-60) - Caution
- **2:** Overbought (RSI > 70) - Sell signal

Permissions Required

- `SYSTEM_ALERT_WINDOW` - Display overlay elements
- `FOREGROUND_SERVICE` - Run background analysis
- `MEDIA_PROJECTION` - Capture screen content

- `RECORD_AUDIO` - Play sound alerts
- `INTERNET` - Data connectivity (optional)

Troubleshooting

No signals detected

- Ensure trading chart is visible on screen
- Check that chart uses standard candlestick colors (green/red)
- Increase sensitivity in Settings

Screen capture not working

- Grant screen recording permission when prompted
- Disable any screen overlay apps
- Restart the application

Sound not playing

- Check device volume settings
- Enable sound alerts in Settings
- Verify microphone permission is granted

Development

Technology Stack

- **Language:** Java 11
- **Min SDK:** 26 (Android 8.0)
- **Target SDK:** 34 (Android 14)
- **Build System:** Gradle 8.1.0
- **Key Libraries:**
 - AndroidX AppCompat
 - MediaProjection API
 - OpenCV (for image processing)
 - ML Kit Vision (for text recognition)

Project Structure

Plain Text

```
eagle-trading-analyzer/  
├── app/  
│   ├── src/main/  
│   │   ├── java/com/eagle/trading/analyzer/  
│   │   │   ├── core/  
│   │   │   │   └── CandleAnalyzer.java  
│   │   │   ├── service/  
│   │   │   │   ├── ScreenCaptureService.java  
│   │   │   │   └── AnalysisService.java  
│   │   │   └── ui/  
│   │   │       ├── MainActivity.java  
│   │   │       └── SettingsActivity.java  
│   │   └── res/  
│   │       ├── layout/  
│   │       ├── drawable/  
│   │       ├── values/  
│   │       └── mipmap/  
│   └── AndroidManifest.xml  
├── build.gradle  
├── build.gradle  
├── settings.gradle  
└── README.md
```

Building APK

Debug APK

Bash

```
./gradlew assembleDebug  
# Output: app/build/outputs/apk/debug/app-debug.apk
```

Release APK (Optimized)

Bash

```
./gradlew assembleRelease  
# Output: app/build/outputs/apk/release/app-release-unsigned.apk
```

Signed Release APK

Bash

```
./gradlew assembleRelease -Pandroid.injected.signing.store.file=keystore.jks \
-Pandroid.injected.signing.store.password=password \
-Pandroid.injected.signing.key.alias=alias \
-Pandroid.injected.signing.key.password=password
```

Performance Optimization

- Frames captured every 2 seconds (configurable)
- Efficient bitmap processing with color-based detection
- Background service prevents UI freezing
- ProGuard optimization for release builds

Future Enhancements

- ☐ Support for multiple timeframes
- ☐ Historical data analysis
- ☐ Custom alert thresholds
- ☐ Export trading history
- ☐ Integration with trading APIs
- ☐ Machine learning pattern recognition
- ☐ Multi-language support

License

This project is provided as-is for educational and personal trading analysis purposes.

Support

For issues, feature requests, or improvements, please refer to the troubleshooting section or contact support.

Version: 1.0.0

Last Updated: May 2026

Minimum Android: 8.0 (API 26)